Industrial Internship Report on

" Automotive Parts E-Commerce System"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was "Automotive Parts E-Commerce System", a full-stack web application developed using Flask, HTML, CSS, and SQLite. The system allows users to register, log in, browse automotive spare parts, add products to a cart, and manage their cart before checkout. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1 Preface

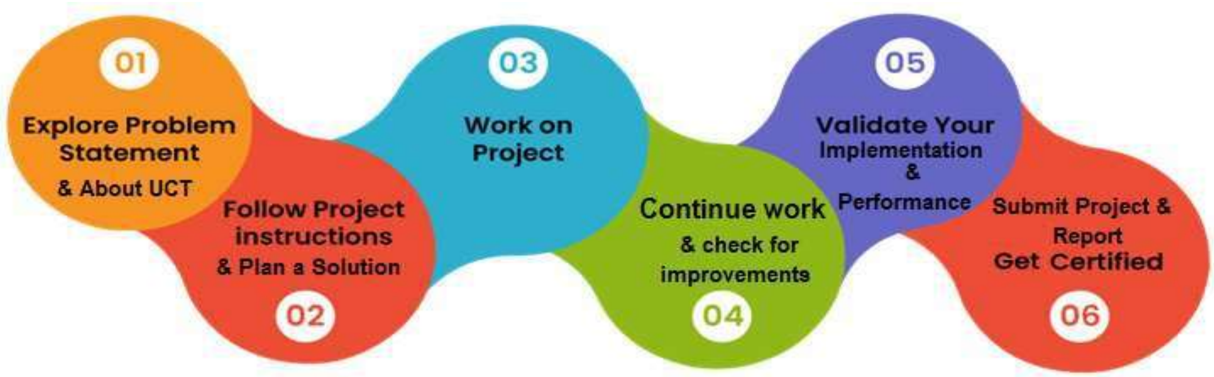
This report summarizes the work carried out during the 6-week Industrial Internship program provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). The internship was designed to provide practical exposure to real-world software development problems and enhance technical and professional skills.

The need for relevant internships plays a crucial role in career development, as it bridges the gap between academic knowledge and industrial requirements. This internship provided hands-on experience in full-stack web development and helped in understanding how real-time applications are designed, developed, and deployed.

My project was "Automotive Parts E-Commerce System", a full-stack web application developed using Flask, HTML, CSS, and SQLite. The system enables users to register, log in, browse automotive spare parts, add products to a cart, and manage their cart efficiently.

The internship program was well-structured and planned over 6 weeks, where each week focused on specific learning objectives, including requirement analysis, frontend development, backend integration, database management, and final project completion.

This internship provided valuable learning experiences in web development, database handling, session management, and problem-solving. It also improved my understanding of full-stack development workflows and real-world project implementation.



I would like to express my sincere gratitude to upskill Campus, The IoT Academy, and UniConverge Technologies Pvt Ltd (UCT) for providing this opportunity. I also thank all mentors and coordinators who guided me throughout the internship journey.

Finally, I would like to dedicate this experience to my peers and juniors, encouraging them to actively participate in internships as they provide essential industry exposure and significantly contribute to career growth.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

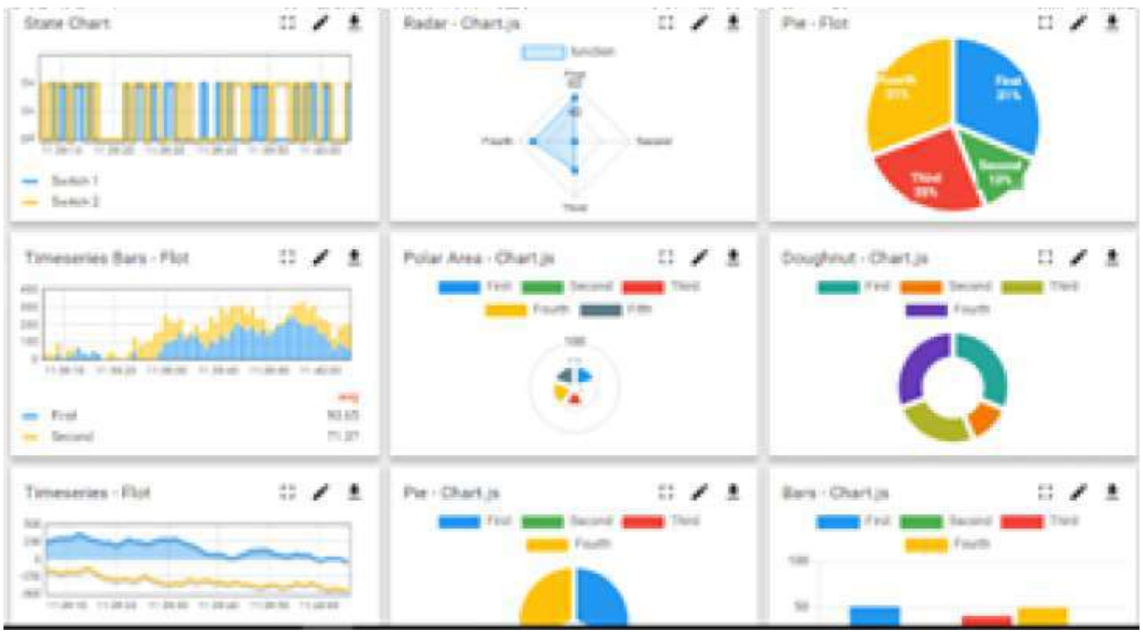
i. UCT IoT Platform ( )

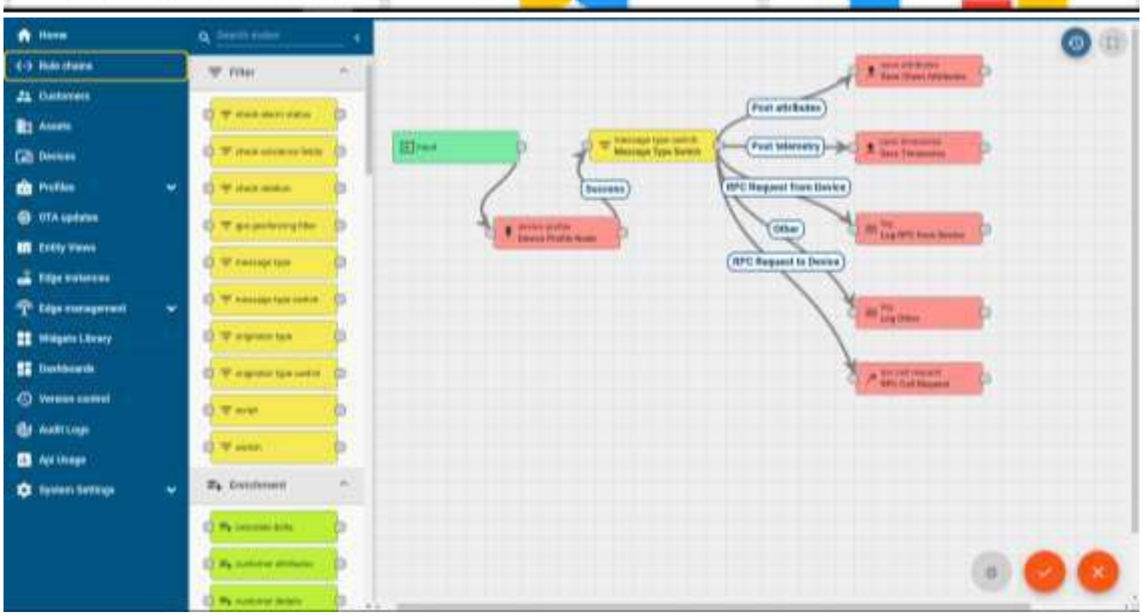
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine





ii. **Smart Factory Platform (FACTORY WATCH)**

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.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Demo
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CNC_S7_B4
Utilization


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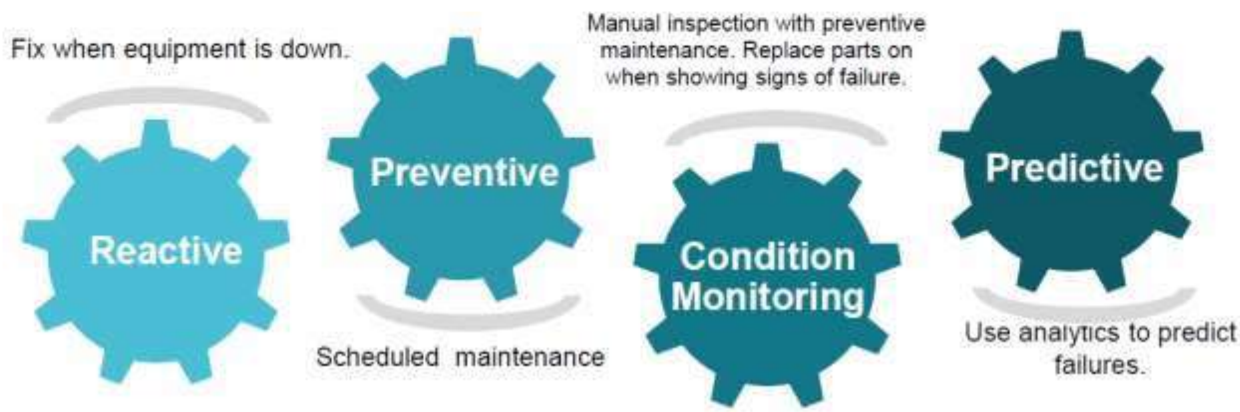
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

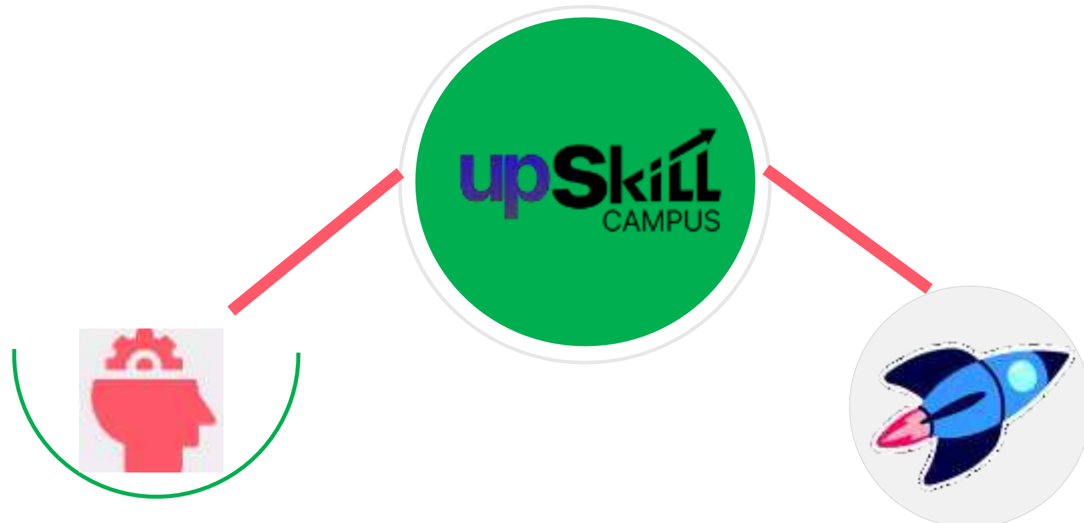
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

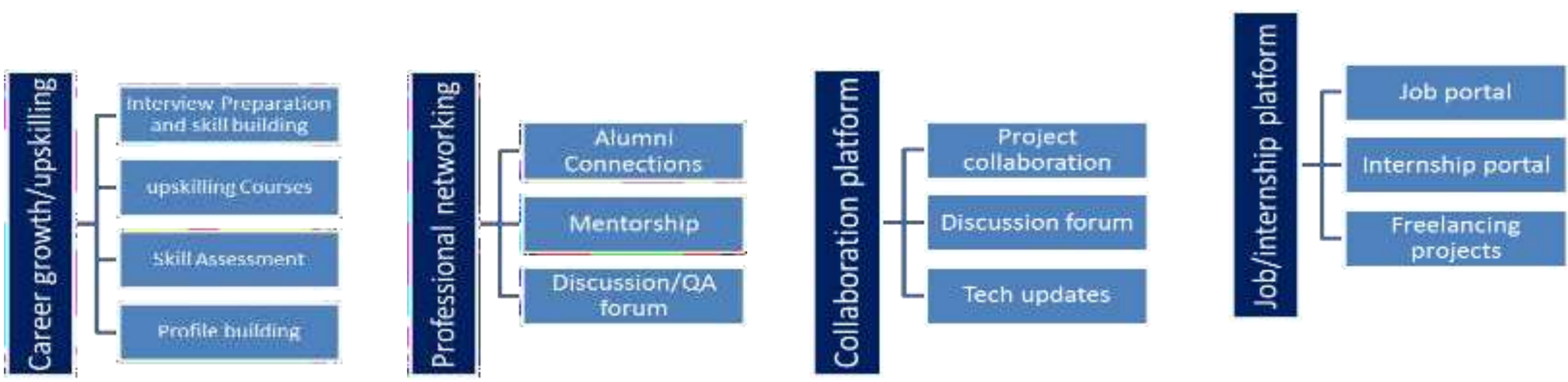
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] UniConverge Technologies Pvt Ltd – Internship Project Guidelines

[2] UpSkill Campus Internship Training Materials

[3] Official documentation and online resources related to Flask, HTML, CSS, and SQLite used during project development

2.6 Glossary

Terms	Acronym
Internet of Things	**IoT**
Flask	Micro Web Framework in Python
F**ull Stack Development**	F**SD**

3 Problem Statement

The assigned problem statement is to design and develop a full-stack web-based application that addresses a real-world business requirement in an efficient and user-friendly manner. In many existing systems, manual processes or outdated software solutions lead to inefficiencies such as data redundancy, slow processing, lack of accessibility, and difficulty in maintaining accurate records. These challenges highlight the need for a modern, scalable, and automated solution.

The objective of this project is to analyze the given requirements and develop a complete web application that integrates frontend, backend, and database technologies. The system should provide a smooth user experience while ensuring proper data management, security, and reliability. It is designed to minimize manual effort and improve the overall efficiency of the process.

The project is implemented using Flask as the backend framework, HTML and CSS for frontend design, and SQLite as the database system. This combination ensures lightweight deployment, easy integration, and fast performance suitable for academic and small-scale industrial applications.

The developed system demonstrates core full-stack development concepts such as user interface design, server-side processing, database connectivity, and CRUD (Create, Read, Update, Delete) operations. It also helps in understanding how real-world applications are structured and deployed.

Overall, this project aims to bridge the gap between theoretical knowledge and practical implementation by simulating an industry-level software development environment. It enhances technical skills, problem-solving ability, and understanding of complete software development lifecycle.

4 Existing and Proposed solution

In the current scenario, several basic web-based and manual systems are available for managing similar business processes. These existing solutions often provide limited functionality such as basic data entry, static interfaces, and minimal automation. While they may fulfill simple requirements, they lack scalability, efficiency, and advanced features required for modern applications.

Limitations of Existing Solutions

The existing systems have several limitations such as:

- Lack of proper automation, leading to increased manual effort
- Poor user interface and user experience design
- Limited scalability for handling large datasets
- No proper integration between frontend and backend systems
- Inefficient data management and retrieval processes
- Lack of security and validation mechanisms

Proposed Solution

To overcome these limitations, a full-stack web application has been proposed. The system is designed using Flask as the backend framework, HTML and CSS for the frontend interface, and SQLite as the database for efficient data storage and retrieval.

The proposed solution focuses on:

- Providing a user-friendly and responsive interface
- Automating data handling processes
- Ensuring efficient CRUD (Create, Read, Update, Delete) operations
- Improving system performance and reliability
- Maintaining structured and secure data storage

This system offers a more modern and scalable approach compared to traditional or basic existing solutions.

4.1 Code submission (Github link)

https://github.com/keerthibandari167/upskillcampus

4.2 **Report submission (Github link)** : first make placeholder, copy the link.

https://github.com/keerthibandari167/upskillCampus

5 Proposed Design/ Model

The proposed system follows a simple full-stack architecture where the frontend communicates with the backend server, and the backend interacts with the database to perform all required operations. The system is designed in a modular way to ensure scalability, maintainability, and easy understanding of the workflow.

The overall design includes three main layers:

- **Frontend Layer**: Responsible for user interaction using HTML and CSS
- **Backend Layer**: Handles business logic using Flask framework
- **Database Layer**: Stores and retrieves data using SQLite

The system flow begins from user input, passes through validation and processing at the backend, and finally stores or retrieves data from the database before displaying the result to the user.

5.1 High Level Diagram (if applicable)

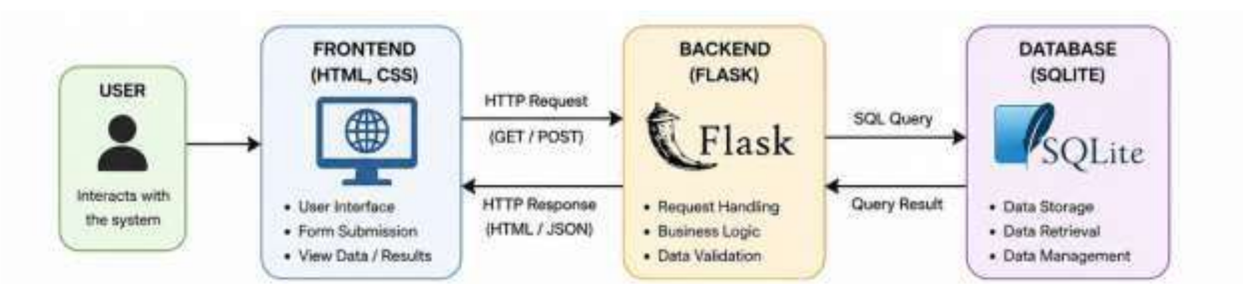
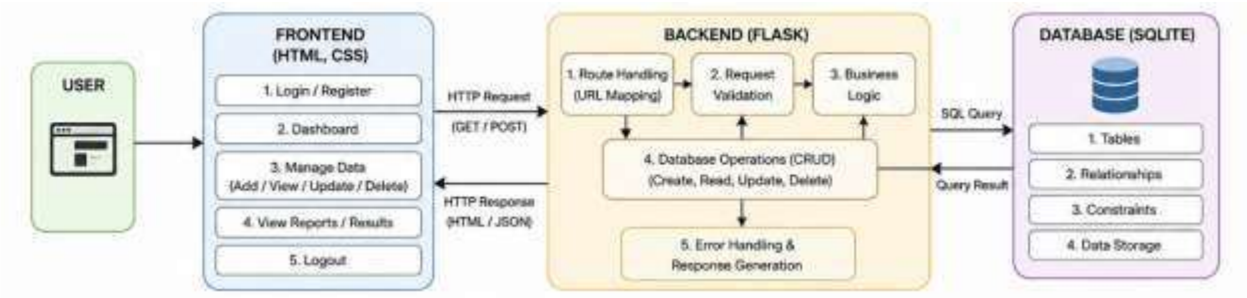
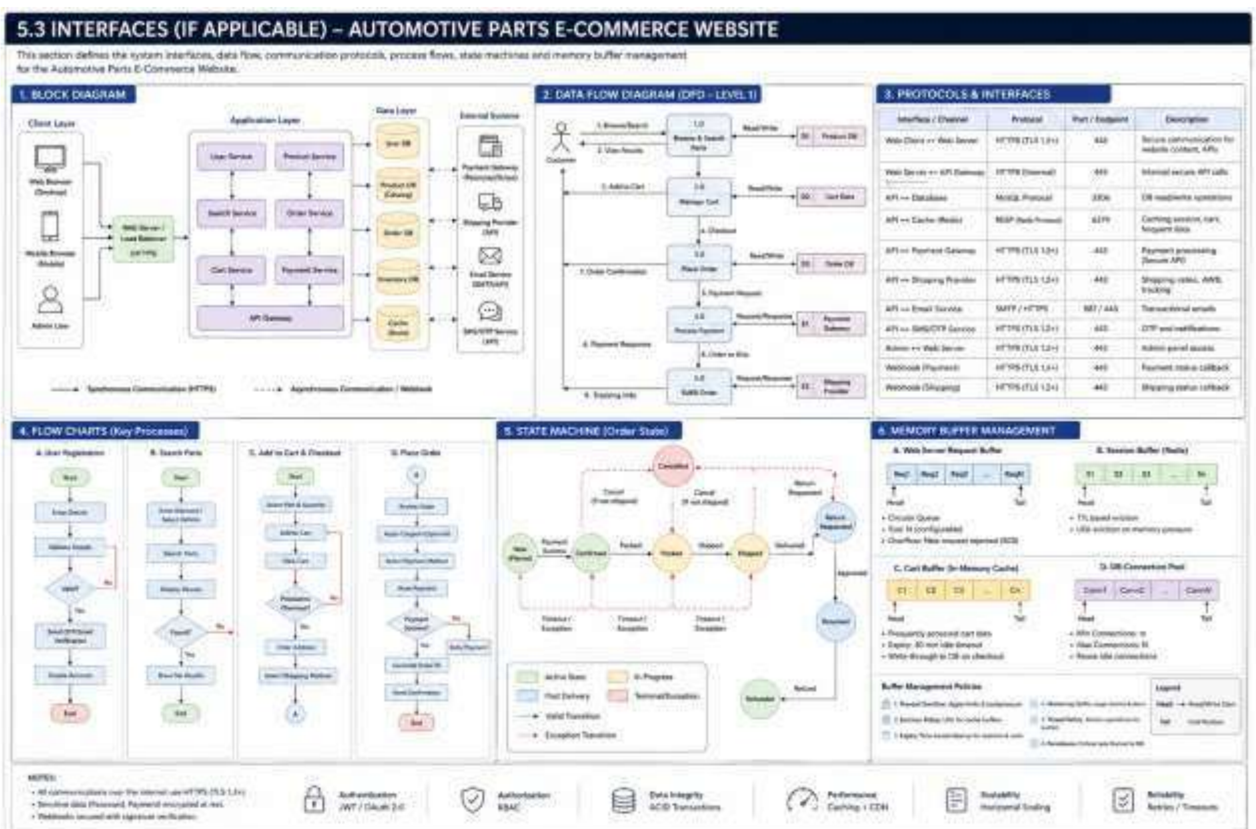


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)



6 Performance Test

This project is designed considering basic performance constraints such as response time, data handling capacity, and system scalability. The system ensures efficient handling of user requests with minimal delay by using lightweight frameworks like Flask and SQLite.

Constraints Considered

- Limited server processing time
- Efficient database query handling
- Minimal memory usage for lightweight deployment
- Fast response time for user requests

These constraints were addressed by using optimized database queries, structured backend logic, and minimal frontend complexity.

6.1 Test Plan/ Test Cases

Test Case	**Input**	**Expected Output**	**Result**
User Login	Valid credentials	Successful login	Pass
Add Record	Valid data input	Data stored in DB	Pass
Update Record	Existing ID	Updated data saved	Pass
Delete Record	Valid ID	Record removed	Pass

6.2 Test Procedure

The testing was performed by executing different functional modules of the application. Each feature was tested individually, including form submission, database insertion, update operations, and deletion operations. Manual testing was conducted to verify correctness of output.

6.3 Performance Outcome

The system performed efficiently under normal usage conditions. All operations such as data insertion, retrieval, update, and deletion were executed successfully without errors. The response time was minimal due to lightweight architecture, ensuring smooth user experience.

7 My learnings

During this internship project, I gained practical experience in full-stack web development. I learned how frontend, backend, and database components interact in a real-world application. I also improved my understanding of Flask framework, HTML/CSS design, and SQLite database operations.

In addition, I developed problem-solving skills, debugging techniques, and learned how to structure a complete software project. This experience helped me bridge the gap between theoretical knowledge and practical implementation, improving my confidence for industry-level projects.

8 Future work scope

In the future, this project can be enhanced by adding advanced features such as:

- User authentication with role-based access control
- Cloud database integration for scalability
- Mobile responsive UI improvements
- Integration of payment gateway (if applicable)
- Advanced analytics and reporting dashboard
- Deployment on cloud platforms like AWS or Azure

These improvements will make the system more robust, scalable, and suitable for real-world industrial applications.